

**Modeling Spatial Dependence: A Simulation-Based Comparison of
Parametric and Semi-Parametric Approaches**

by

Your Name Here

A thesis submitted in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE
in
ELECTRICAL ENGINEERING

UNIVERSITY OF PUERTO RICO
MAYAGÜEZ CAMPUS
20XX

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ABSTRACT

Hi! We encourage you to visit <https://libguides.uprm.edu/writingclinics> and check out the **Abstracts Clinic**. Keep in mind that depending on your discipline, abstracts should be a **single paragraph**, at around **200-400 words**. It should concisely but clearly summarize your thesis document. The **IMRaD format** is recommended for writing abstracts: Introduction (1-3 sentences long, present tense), Methodology (1-3 sentences long, past tense), Results (1-3 sentences long, past tense), and Discussion (1-2 sentences long, present tense). Remember that the number of sentences and verb tense are only guidelines!

RESUMEN

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List of Acronyms

PR	Puerto Rico
UPR	Universidad de Puerto Rico
RUM	Recinto Universitario de Mayaguez
GRIC	Graduate Research and Innovation Center
GWF	Graduate Writing Facilitator

List of Symbols

kg	kilogram
τ	tau
μL	microliters

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2026

Chapter 1

INTRODUCTION

1.1 Motivation, Purpose, Justification

Hi! We encourage you to visit <https://libguides.uprm.edu/writingclinics>. The GWF Writing Clinics provide in-depth, useful information for preparing: abstracts, literature reviews, citations (Mendeley), academic writing, thesis outline, grammatically-sound writing, presentations (communication strategies in oral presentations and poster sessions) and visual design.

1.2 Objectives, Research Questions, Hypothesis

Example of bullets. Mention objectives, research questions, or hypotheses:

- Analyze
- Study
- Identify
- Construct

- Analyze

Example of description. The work is divided in three phases:

Phase I: Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis.

Phase II: Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna.

Phase III: Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo.

1.3 Chapter Summary

This thesis consists of 6 chapters which are briefly summarized below:

Chapter 1: Serves as an introduction to the investigation.

Chapter 2: Presents an overview of Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo.

Chapter 3: Provides a means of presenting the Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo.

Chapter 4: Discusses what a Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo.

Chapter 5: Presents a simulation model for Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo.

Chapter 6: Presents Results for Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo.

Chapter 7: Makes final remarks regarding the study and proposes several ideas for future work.

Chapter 2

Sections and Subsections

2.1 How to add sections/subsections to your $\text{\LaTeX}$ document

$\text{\LaTeX}$ provides three levels of nested subsections by using the commands **section**, **subsection** and **subsubsection**.

2.2 Section 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit.

2.2.1 Subsection 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit.

2.2.1.1 Subsubsection 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Chapter 3

Citations, Images, and Equations

You can visit <https://en.wikibooks.org/wiki/LaTeX> for an in-depth guide to the L^AT_EX typesetting system.

3.1 How to incorporate citations in your document

Make sure you have your references file in **.bib format**. You can export it from Mendeley and copy/paste them into the **referencias.bib** file. Using the **cite command**, start writing your source identifier and Overleaf will automatically show you the available references. A sample citation [1], and another citation [2], and two citations [3],[4] are shown as examples.

3.1.1 Citation Styles

There are several bibliography styles to choose from: **abbrv**, **acm**, **alpha**, **apalike**, **ieeetr**, **plain**, **siam**, **unsrt**. The default is **ieeetr**. Remember to change the option in the document preamble in **thesis.tex**

3.2 Using Images

This is an example of a figure, as shown in Figure 3.1. Your images must be uploaded to the **images** folder. Accepted file formats are **pdf**, **png**, **jpg** and **eps**. Use the following options for the location of the image on the page: **[htbp]** that refer to here, top, bottom, or special page. Pay attention to the image width. If the image is wider than the margins, L^AT_EX will not compile. Remember that you must make a reference to your figure or table BEFORE it appears in the document.



Figure 3.1: GRIC logo!!

3.2.1 Subfigures

If you need to place two subfigures in your figure, follow the example below:



(a) Put your sub-caption here



(b) Put your sub-caption here

If you need to place four subfigures in your figure, follow the example below, but L^AT_EX is very particular with widths, so you have to play around with the numbers. It might

give you overflow warnings. These won't stop your document from compiling. It is easier to build the four images as a single one before uploading it to Overleaf.



(a) Put your sub-caption here



(b) Put your sub-caption here



(c) Put your sub-caption here



(d) Put your sub-caption here

Figure 3.3: Put your caption here

You can also plot graphs in $\text{\LaTeX}$ using Pgfplots. This is a visualization tool to make it simpler to include plots in your documents. The basic idea is that you provide the input data/formula and pgfplots does the rest. You are welcome to experiment with this package, but really, but please, just use excel and export the image! A sample graph is shown in Figure 3.4.

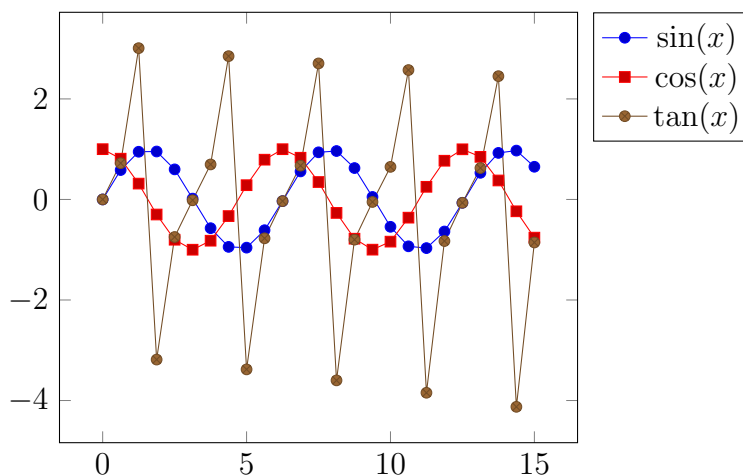


Figure 3.4: Put your Caption

3.3 Typing Equations in L^AT_EX

Writing equations is a bit complicated at first. It will become second nature once you get used to the symbols used. Equations are written between **begin equation** and **end equation** commands, and are referenced using `theref` command and the equation label, as in equation 3.1:

$$\int_0^5 \frac{2}{3}x \, dx \quad (3.1)$$

You have several options for incorporating your equations in L^AT_EX if you don't want to type them in the editor. Equation 3.1 was typed in MS Word Equation Editor and copy/pasted here. You can also use online equation converter tools, such as: <https://latex.codecogs.com/eqneditor/editor.php>.

$$\int_0^\infty x^2 \, dx \quad (3.2)$$

3.3.1 Subsection

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3.3.2 Subsection

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Chapter 4

Tables in L^AT_EX

4.1 How to Incorporate Tables

Tables can be typed directly in the editor, but it's tedious, time-consuming, and can lead to errors. The easiest way to format a table to use in L^AT_EX is to simply type it in MS Excel and convert to L^AT_EX using **Excel2L^AT_EX**: <https://ctan.org/tex-archive/support/excel2latex>. Download the zip file, unzip, run the .xla file and you're ready to go! Type your data on MS Excel, go to Add-ins, click on the Excel@L^AT_EX icon. A window with some options will appear. Click on 'Copy to clipboard' and paste it in your L^AT_EX document. You must enable macros for the add-in to work. Be sure to format the table as you want it to appear in your document. You can reference tables the same way as images, but using the **ref command** and **tab**, as seen in Table 4.3.

Table 4.1: A centered random table

Material	Weight (lb)	Height (ft)
1	4.5	2
2	3.1	3
3	6.8	4.5
4	9.8	10

Including centering and borders/lines.

Table 4.2: This looks weird

Material	Weight (lb)	Height (ft)
1	4.5	2
2	3.1	3
3	6.8	4.5
4	9.8	10

That looks bad, so sub the top, mid and bottomrule for hline.

Table 4.3: This is better!

Material	Weight (lb)	Height (ft)
1	4.5	2
2	3.1	3
3	6.8	4.5
4	9.8	10

Or use thin lines instead.

Table 4.4: This is best!

Material	Weight (lb)	Height (ft)
1	4.5	2
2	3.1	3
3	6.8	4.5
4	9.8	10

4.1.1 Bigger Tables

For tables that don't fit the width of the paper, use the **landscape** command, as shown below:

Tables may be too big, so you can use footnotesize to make the font smaller, like in Table 4.5.

Table 4.5: You can use the footnotesize command to make the font smaller

Temp (C)	Composition (mole%)	Tf (C)	Heat of Fusion (kJ/kg)	Relative Cost
367	NaOH, NaCl (20)	370	370	1.001
	KOH	360	167	4.39
347	KNO ₃ , KBr (10), KCl (10)	342	140	4.04
	NaCl, KCl (24), LiCl (43)	346	281	5.44
328	KNO ₃	337	116	3.8
	KNO ₃ , KCl (6)	320	150	3.33
	NaOH	318	158	2.78
		286-299	3162	1.2
308	NaCl, NaOH (93.7)	314	-	-
	NaNO ₃	310	174	1.35
	NaF, NaNO ₃ (96.5)	304	-	-
	LiOH, KOH (60)	314	341	3.57
289	Na ₂ SO ₄ , NaCl (8.4), NaNO ₃ (86.3)	287	176	1.3
	NaNO ₃ , NaCl (6.4)	284	171	1.2
	KNO ₃ , Ba(NO ₃) ₂ (87)	290	124	2.85
	NaNO ₂	282	212	3.33
	NaNO ₂ , KNO ₃ (45.2)	285	152	3.61
	NaOH, NaCl (7.8), Na ₂ CO ₃ (6.4)	282	316	2.9

Chapter 5

Literature Review

For tips and guidelines on how to write your Literature Review, visit the GWF clinic section at: <https://libguides.uprm.edu/writingclinics>, Clinics 2020.

5.1 Section

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5.1.1 Subsection

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5.1.1.1 Subsubsection

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5.1.2 Subsection

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5.1.2.1 Subsubsection

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5.2 Section

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5.2.1 Subsection

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eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla.

Chapter 6

Methodology

6.1 Section

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6.1.1 Subsection

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6.1.2 Subsection

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Chapter 7

Results

7.1 Section

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7.1.1 Subsection

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Chapter 8

Conclusions

8.1 Section

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APPENDICES

Appendix A

MATLAB Code

Appendix A goes here.

A.1 Section

Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque.

A.2 Section

Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque.

A.2.1 Subsection

Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque.

Appendix B

Database

Appendix A goes here.

B.1 Section

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B.2 Section

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B.2.1 Subsection

Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque.

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